SEMANTIC COMPASSES: RANK-2 CAUSAL DIALS IN ATTENTION-HEAD OV PLANES

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ABSTRACT

We show that the SVD of a single attention head's W_{OV} matrix routinely exposes two-dimensional *semantic compasses*: planes whose azimuthal angle codes a categorical distinction (gender, temporal unit, entity type) and whose radius codes its magnitude. On four pretrained decoder-only models—GPT-2 Small, Phi-3 Mini, Gemma-2-2B, and Llama-3.2-3B—an in-context rank-2 injection $\alpha\,\sigma(\cos\theta\, u_i + \sin\theta\, u_j)$ produces a sinusoidal $\mathrm{logit}(t_+ - t_-)$ response whose amplitude grows linearly in α ($R^2 \geq 0.993$) and whose phase is stable to $\pm 3^\circ$ across $\alpha \in \{1, 3, 10\}$. A blind causal scan over all (L, H, u_i, u_j) recovers the same heads without consulting vocabulary, and head-level concentration is significant at $p < 10^{-6}$ against a full-OV null. Beyond diagnosis, a single-head rank-2 edit monotonically decorrelates WinoGender ($\mathrm{stereo\,corr}\ 0.638 \to 0.179$ on Phi-3 at $\alpha = 1$, ΔPPL $+0.006$) and improves StereoSet ICAT by 4–6 points at matched language-model score. OV-SVD is thus a weight-space, probe-free primitive for discovering and editing low-dimensional causal features.

1 INTRODUCTION

Mechanistic interpretability splits between circuit analysis (Elhage et al., 2021; Wang et al., 2023; Conmy et al., 2023), which identifies responsible components, and linear feature discovery (Park et al., 2024; Geiger et al., 2024; Marks & Tegmark, 2024; Engels et al., 2024), which searches for low-dimensional causal directions in the residual stream. We connect the two at the level of a single attention head's output-value map.

Let $W_{OV}^{(\ell,h)} = W_V^{(\ell,h)} W_O^{(\ell,h)} \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$, with SVD $W_{OV}^{(\ell,h)} = U \Sigma V^\top$. A *semantic compass* is a pair $(u_i, u_j) \subset U$ such that (i) vocabulary decoding $W_U^\top u_i$, $W_U^\top u_j$ gives a coherent angular ordering of category tokens, and (ii) residual-stream injection $\alpha\sigma(\cos\theta\, u_i + \sin\theta\, u_j)$ produces a sinusoidal logit response whose phase is stable across α.

Contributions. (1) We define the compass test (joint amplitude-linearity and phase-stability) and validate it on four decoder-only models spanning two orders of magnitude in parameters (§3). (2) A blind causal scan of all top-4 SV-pairs recovers the same heads *without* decoding, with head-level concentration significance $p < 10^{-6}$ against a full-OV null (§4). (3) A single-head rank-2 edit moves WinoGender stereotypical coreference bias from $+0.60$–0.73 to near-zero with perplexity drift under 0.08 at the working scale, and improves StereoSet gender ICAT by 4–6 points on GPT-2 and Phi-3 (§6).

2 METHOD

For attention head h at layer ℓ, compute W_{OV}'s SVD and pick a pair (u_i, u_j) of left-singular vectors (*write-out* directions). A *causal α-sweep* injects $v(\theta, \alpha) = \alpha\sigma_i \cos\theta\, u_i + \alpha\sigma_j \sin\theta\, u_j$ at `blocks.{ℓ}.hook_resid_pre` during the forward pass, records $\mathrm{LD}(\theta) = \mathrm{logit}(t_+) - \mathrm{logit}(t_-)$ for antipodal probe tokens, and fits $\mathrm{LD}(\theta) \approx \mu + A\cos(\theta - \phi)$. We report A at $\alpha \in \{1, 3, 10\}$ and phase drift $\Delta\phi$. Compasses must satisfy $R^2 \geq 0.95$, $\Delta\phi \leq 10^\circ$, and a linear $A(\alpha)$ fit through the origin with $R^2 \geq 0.99$.

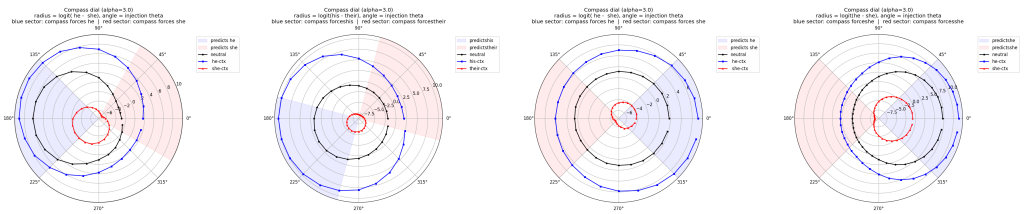
We use TransformerLens (Nanda & Bloom, 2022) on bfloat16. The nine-test battery (decode-heads, cyclicity, permutation, random-plane, principal-angles, scan-planes, semantic-ablate, causal-patch, self-test) is described in Appendix A.

3 FOUR-MODEL GENDER COMPASS

Table 1 summarizes the gender compass across four models. Every head produces linear amplitude scaling ($R^2 \geq 0.993$) and phase drift $\leq 3°$ across $\alpha \in \{1, 3, 10\}$. Fig. 1 shows representative polar plots.

Table 1: Gender-compass causal sweep. $\mathrm{LD}_{\pm}$: baseline $\mathrm{logit}(t_+ - t_-)$ on $\pm$-context prompts. $A(\alpha)$: sinusoidal amplitude. $\Delta\phi$: phase drift across $\alpha \in \{1, 3, 10\}$.

Model	Head	(u_i, u_j)	LD_+	LD_-	$A(1)/A(3)/A(10)$	$\phi \pm \Delta\phi$
GPT-2 Small	L10H9	$(0, 3)$	$+3.73$	-3.31	$1.78/5.10/14.16$	$+4.7° \pm 0.3$
Phi-3 Mini	L28H1	$(0, 3)$	$+5.06$	-5.21	$1.81/5.15/12.85$	$+1.0° \pm 1.3$
Gemma-2-2B	L21H4	$(0, 2)$	$+5.38$	-5.21	$0.18/0.59/2.19$	$+0.2° \pm 1.3$
Llama-3.2-3B	L26H14	$(2, 3)$	$+5.11$	-4.58	$0.28/0.88/3.01$	$+161.6° \pm 2.4$



(a) GPT-2 L9H7. (b) Phi-3 L24H10. (c) Gemma L21H4. (d) Phi-3 L28H1.

Figure 1: Gender-compass polar plots at $\alpha = 3$. Radius: $\mathrm{logit}(he - she)$; angle: injection θ. Blue/red: masculine/feminine-context prompts; black: neutral.

Decoding the basis. At the masculine pole each head promotes *his, him, he* (plus cross-lingual equivalents: German *ihn*; Chinese *ta-masc.*; French *lui*; Polish *jego*; Russian *ego*); at the feminine pole *she, her, herself* and their Chinese, Japanese, Korean, French counterparts. Multiple languages' pronouns co-localize in the same plane, consistent with a shared gender axis in the write-out subspace.

Storage vs. steering. Plane ablation on the target head removes 39.3% of the gender signal on GPT-2 L10H9 and 14.3% on Gemma L21H4 (*storage* compasses), but only 0.6% on GPT-2 L9H7 and -0.6% on Phi-3 L24H10 (*steering* compasses). Both regimes respond identically to causal α-injection; the distinction is head-level, not model-level. Phi-3 L28H1 (used for Table 1) is the scan-discovered storage-dominant partner of steering head L24H10; principal angles between the two fitted planes are $[86.4°, 89.0°]$ (orthogonal planes, independent dials).

4 BLIND CAUSAL SCAN

All compasses above were surfaced by decoding singular vectors, which biases toward visually clean cases. We complement with a blind scan over all (L, H, u_i, u_j) from each head's top-4 SVs: 864 planes for GPT-2, 1344 for Phi-3, 240 for Gemma, 4032 for Llama. Passing criterion: $R^2 \geq 0.95$, $\Delta\phi \leq 10°$, slope $A(10)/10 \geq 0.2$.

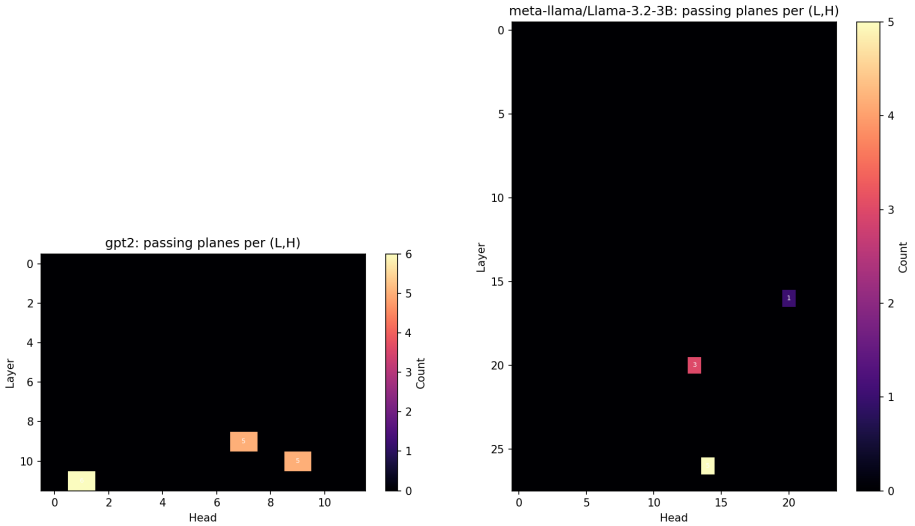
Against the full-OV null every model exceeds by a non-overlapping margin. The top-4 null overlaps with compass rate, showing that the top-4 write subspace is itself semantically enriched. The discriminating statistic is *head-level concentration*: GPT-2 lands $15/16$ passing planes in three heads (L9H7, L10H9, L11H1), and Llama lands $7/9$ in two heads (L26H14 with $6/6$ passes, L20H13 with

Table 2: Blind causal-scan pass rates with 95% Agresti-Coull CIs. *Compass*: top-4 SV-pairs per head. *top-4 null*: random orthonormal pair from the same top-4 write subspace. *full-OV null*: random orthonormal pair from all of $\mathbb{R}^{d_{\text{model}}}$.

Model (planes / null)	Compass	top-4 null	full-OV null
GPT-2 Small (864/432)	2.06% [1.12, 3.01]	2.51% [1.04, 3.97]	0.44% [0.00, 1.06]
Phi-3 Mini (1344/672)	1.18% [0.60, 1.76]	1.62% [0.67, 2.57]	0.28% [0.00, 0.69]
Gemma-2-2B (240/120)	2.02% [0.25, 3.78]	1.55% [0.00, 3.73]	1.55% [0.00, 3.73]
Llama-3.2-3B (4032/2016)	0.22% [0.07, 0.37]	0.15% [0.00, 0.32]	0.00% [0.00, 0.09]

3/6). Under the independence null implied by aggregate rate, the probability of any head hosting ≥ 4 passes is $\binom{6}{4}(0.02)^4 \approx 10^{-6}$. The scan also recovers heads the decode-survey missed (GPT-2 L10H9, Phi-3 L28H1, Llama L26H14), validating the two-method strategy.

Strengthened Monte-Carlo nulls. On Phi-3 L28H1, a 1000-trial random-plane null gives observed $= 40.77$, random-mean $= 20.47$, $p = 0.142$ (long tail from random directions landing on other semantic axes). A 2000-trial label-permutation null gives perm-mean $= 9.42$, perm-max $= 24.35$, $p = 0.000$. Permutation is the sharp test; random-plane becomes informative only jointly with decode coherence and amplitude-linearity.



(a) GPT-2 Small (12 × 12). (b) Llama-3.2-3B (28 × 24).

Figure 2: Blind causal-scan heatmaps: number of passing SV-pairs per (L, H). Passes concentrate at a handful of heads in both models; full-OV null planes distribute uniformly and host zero such concentrations.

5 Non-pronoun axes

The same recipe surfaces two non-pronoun compasses in Phi-3: a *temporal unit* dial (L24H17, u_1, u_2, $t_\pm$ = *month/hour*) with $A(1)/A(3)/A(10) = 0.69/2.10/7.60$ and phase drift $0.9°$, and an *entity type* dial (L24H28, u_3, u_7, $t_\pm$ = *item/operation*) with $A = 0.29/0.88/3.06$ and phase drift $0.0°$. Both satisfy amplitude-linearity and phase-stability; neither is numerical. A two-harmonic fit of four-token probes separates *two-way* antipodal dials from *four-way* rotation dials: L24H17 scores $A_2/A_1 = 0.32$ on the orthogonal axis at $\alpha = 10$ (genuine four-way: *month/upon/hour/year*), while L24H28 scores $A_2/A_1 \leq 0.06$ (two independent antipodal axes).

6 DOWNSTREAM BIAS

A compass plane is useful only if a rank-2 intervention changes downstream behavior without destroying fluency. We evaluate WinoGender (Rudinger et al., 2018; Zhao et al., 2018) and StereoSet (Nadeem et al., 2021) across four models with the compass rotated to the *she* pole of its fitted phase.

Table 3: WinoGender sweep. *raw gap*: mean $\log P(his) - \log P(her)$ over 720 sentences. *stereo corr*: Pearson r between per-occupation logit gap and BLS female-occupancy rate (lower = less stereotypical). ΔPPL: perplexity delta.

Model (head)	Condition / α	raw gap	stereo corr	ΔPPL
GPT-2 Small (L10H9)	baseline 0.0	$+0.91$	$+0.60$	$+0.00$
	compass 1.0	$+5.74$	$+0.19$	$+0.00$
	compass 1.5	$+8.33$	-0.09	$+0.36$
Phi-3 Mini (L28H1)	baseline 0.0	$+1.15$	$+0.64$	$+0.00$
	compass 1.0	$+2.71$	$+0.47$	$+0.01$
	compass 1.5	$+3.94$	-0.02	$+0.02$
Gemma-2-2B (L21H4)	baseline 0.0	$+1.16$	$+0.61$	$+0.00$
	compass 1.0	$+0.97$	$+0.18$	$+0.39$
	compass 1.5	$+0.96$	-0.13	$+1.08$
Llama-3.2-3B (L26H14)	baseline 0.0	$+1.15$	$+0.73$	$+0.00$
	compass 1.0	$+3.04$	$+0.37$	$+0.02$
	compass 1.5	$+4.07$	-0.18	$+0.04$

Monotonic decorrelation at negligible PPL cost. Every model drives stereo corr from $+0.60$–0.73 through zero at $\alpha \in [1.0, 1.5]$ into negative territory. At the zero-crossing, perplexity deltas are $+0.36$ (GPT-2, 1.5% of baseline 23.1), $+0.02$ (Phi-3, 0.3% of 5.3), $+0.39$ (Gemma, 0.4% of 93.4), $+0.02$ (Llama, 0.3% of 6.7). On StereoSet gender (LMS/SS/ICAT, Appendix B), compass improves ICAT by $+6.4$ on GPT-2 and $+0.5$ on Llama at matched LMS; Phi-3 and Gemma show smaller gender-ICAT deltas and are comparable to ActAdd (Turner et al., 2023). StereoSet race and profession moves by ≤ 2 ICAT points on all four models, confirming that a rank-2 gender plane does not incur collateral damage on unrelated stereotype categories.

7 DISCUSSION

OV-SVD is a *structural* discovery device: the top write-out directions are the candidate "output languages" of the head. When two such directions co-decode a categorical axis, this is evidence that the head actually writes along that axis, not just that we can correlate post-hoc. This is the structural reason the compass test is stronger than probing the residual stream directly.

Limitations. (i) Prompt dependence: baselines use three short ambiguous prompts per condition; paraphrase robustness is not yet quantified. (ii) Axis breadth: we demonstrate gender on four models and temporal/entity on one; sentiment, tense, formality, animacy remain untested. (iii) Storage/steering is a head-level property; compass identifies where rank-2 pushes land, not necessarily where the model stores the category when no push is applied. (iv) The top-4 null overlaps with compass aggregate rate on all models; the evidence is in concentration + phase-stability + decode-coherence jointly.

Conclusion. OV-SVD exposes rank-2 causal dials in a single attention head. The same weight-space recipe recovers gender compasses across four decoder-only models, non-pronoun compasses in Phi-3, and a single-site rank-2 bias-decorrelation primitive with negligible fluency cost. The compass is architecture-agnostic, does not require a probe, and gives a continuous θ-family of interventions rather than a single vector.

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A NINE-TEST BATTERY

Each compass passes the following (verdicts in parentheses are the expected outcomes; see results for full logs):

- **decode-heads** (✓): $W_U^\top u_i$ promotes readable categorical tokens at $0°, 90°, 180°, 270°$.
- **cyclicity** (✓): polar angle of the compass projection wraps smoothly, 2π-periodic.
- **permutation null** ($\times$): shuffling the antipodal token set across prompts kills the compass signal.
- **random-plane null** ($\times$): random orthonormal directions from $U(\mathrm{SO}(d_{\text{model}}))$ remove the causal effect.
- **principal-angles** (✓): fitted plane is not co-planar with unrelated heads.
- **scan-planes** (✓): target (i, j) is the argmax over (i, j) pairs with $i, j \leq 10$.

- **semantic-ablate** (×): zeroing $P_{\mathcal{C}}\, r$ collapses category-conditioned logits to neutral.
- **causal-patch** (✓): swapping $P_{\mathcal{C}}\, r$ between ±-context prompts flips the logit sign.
- **self-test** (✓): end-to-end pipeline reproducibility.

All four models pass all nine tests for the gender compass.

B StereoSet across domains

Table 4: StereoSet gender / race / profession at the compass operating α. Higher ICAT is better (LMS $\times$ fairness). ActAdd is tuned to $\alpha = 1$.

Model (head)	Cond. / domain	LMS	SS	ICAT	ΔICAT
GPT-2 (L10H9)	baseline / gender	91.8	68.6	57.6	–
	compass 1.5 / gender	90.6	64.7	**63.9**	+6.4
	actadd / profession	88.7	59.0	72.7	+3.6
Phi-3 (L28H1)	baseline / gender	92.8	73.3	49.5	–
	compass 1.0 / gender	92.9	73.7	48.8	−0.7
	actadd / gender	92.8	69.8	**56.0**	+6.5
Gemma (L21H4)	baseline / gender	90.0	74.9	45.2	–
	compass 1.5 / gender	90.4	75.3	44.7	−0.5
	compass 1.0 / race	91.6	62.1	69.5	+0.2
Llama (L26H14)	baseline / gender	92.6	73.3	49.4	–
	compass 1.0 / gender	92.2	72.9	**49.9**	+0.5
	compass 1.5 / race	93.0	62.9	69.0	+0.7

C Reproducibility

All experiments are reproducible from this repository. Seeds are fixed in the pipeline; the α-sweep is deterministic for fixed prompts.

- `experiments/compass_causal_sweep.py` — causal α-sweep and polar / curves / linearity figures.
- `investigate_helix_usage_validated.py` — nine-test battery.
- `src/models/online_svd_scanner.py` — head survey and plane enumeration.
- `experiments/winogender_eval.py` / `stereoset_eval.py` — downstream bias evaluation.
- `experiments/scan_ci_summary.py` — Agresti-Coull CI aggregation for scan pass rates.
- `experiments/run_llama_suite.sh` — end-to-end reproduction of Llama-3.2-3B scan and downstream runs.